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https://app.thinkroot.dev/apps/86164a5d-eb8a-44d3-b10f-62c15fad9faf

ThinkRoot | All Apps | Forensica AI | Save | Support | Feedback | Private Workspace 100 credits

Documents | forensica_ai_docs.sl | EN | Run

Normal text

Forensica AI — Hackathon Documentation Landing Page

Overview

Forensica AI is an explainable AI-powered document verification platform that detects forged or manipulated documents. Visitors should review the prototype/demo and download the verification report or project brief for judging evaluation.

Conversion Goal

- **Primary CTA:** "View Demo & Report" (opens interactive prototype and sample report)
- **Secondary CTA:** "Download Project Brief" (PDF of system design and evaluation)

Page Shell

- **Layout:** Single page, vertical scroll, clearly-separated sections matching judge evaluation checklist
- **Theme:** Professional, high-contrast (dark slate background #0F172A, accent teal #06B6D4), readable sans-serif, ample whitespace, code/plain-text style for technical blocks

Sections

Hero

- **Headline:** Explainable AI for Document Forgery Detection
- **Subheadline:** Forensica AI flags manipulated PDFs, certificates, IDs and images — then explains why with visual overlays and a structured verification report.
- **CTA:** "View Demo & Report"
- **Visual:** Split-screen: left — sample document with highlighted suspicious regions; right — snapshot of verification report (risk score, anomaly list)

Problem/Pain Point

- **Content:**

Describe what you'd like to add, update, or fix in the spec... | Enhance

Fix an issue

Forensica AI | Problem | Solution | Architecture | Pipeline | Tech Stack | View Demo

Track C — Explainable AI for Document Forgery Detection

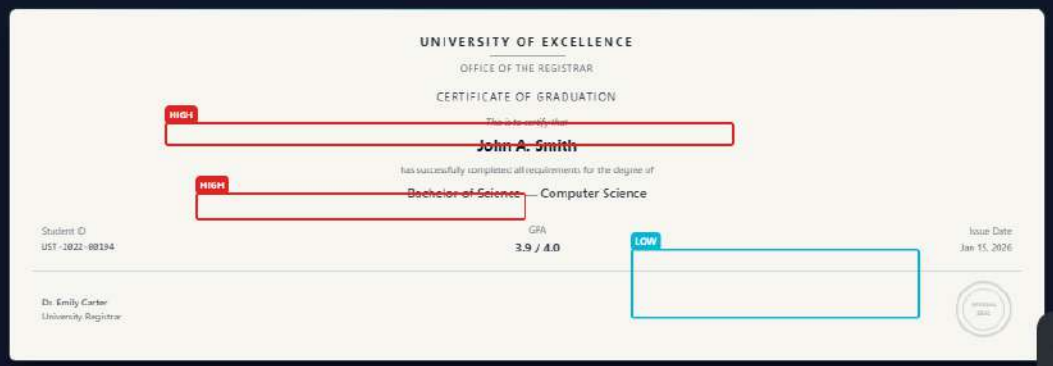
Explainable AI for Document Forgery Detection

Forensica AI flags manipulated PDFs, certificates, IDs, and images — then **explains why** with visual overlays and a structured verification report.

View Demo & Report | Download Project Brief

End-to-end encrypted | 95% detection rate — 92% precision | Live prototype available

DOCUMENT ANALYSIS — LIVE PREVIEW



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Forensica AI

Problem | Solution | Architecture | Pipeline | Tech Stack | View Demo

Track C — Explainable AI for Document Forgery Detection

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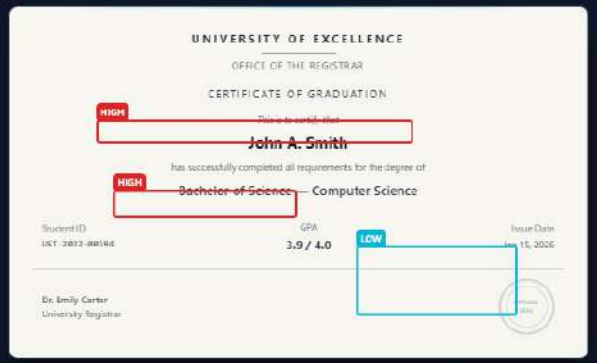

High (2) | Medium (1) | Low (1) | Click boxes for details

VERIFICATION REPORT

DOC ID: doc_7F3k2m9x

university_certificate_sampl...

HIGH RISK

Authenticity Score 34%

DETECTED ANOMALIES (4)

Font Inconsistency HIGH
GPA field uses a different typeface (Helvetica Neue) than the rest of the certificate (Times New Roman). Likely text replacement.
Font Analysis

Pixel Noise Anomaly HIGH
JPEG compression artifacts around the student name differ significantly from surrounding regions — indicative of patch editing.
Pixel Forensics

Metadata Mismatch MED
EXIF creation date (2024-03-10) conflicts with claimed issue date (2026-01-15). Document was likely created before the stated date.
Metadata Forensics

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THE PROBLEM

Document Fraud Is Escalating

Manual verification is slow, inconsistent, and requires specialists that most organizations can't afford to staff at scale.

Education & Admissions

Fraudulent academic certificates and transcripts compromise merit-based admissions, undermining institutional trust.

Hiring & Recruitment

Fake degree certificates and professional credentials slip past manual HR reviews, putting organizations at risk.

Government & Compliance

Forged identity documents and licenses bypass regulatory checks, creating legal and security vulnerabilities.

THE SCALE OF THE PROBLEM

4–8 hrs
average manual review time per document
vs. 2.3s with Forensica AI

$5.4B
annual cost of credential fraud globally
per industry estimates

1 in 8
job applicants misrepresent qualifications
per background screening reports

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THE SOLUTION

Forensica AI — Intelligent Verification

Forensica AI combines OCR, computer vision, and metadata forensics into a modular analysis pipeline. Every flag includes a confidence score and plain-language explanation — aligning directly with **Track C's Explainable AI for Document Forgery Detection**.

ANALYSIS FLOW

Upload
PDF · JPG · PNG

→

Analysis
6-module pipeline

→

Overlays
Visual anomaly map

→

Report
Risk score + JSON

Multi-Format Support

Accepts PDFs, JPG, and PNG. Handles certificates, IDs, invoices, and more.

OCR + Computer Vision

Tesseract OCR extracts text structure; OpenCV detects pixel-level manipulation.

Font & Layout Analysis

Identifies mixed typefaces, spacing anomalies, and alignment inconsistencies.

Explainable Reports

Region-level overlays with confidence scores and plain-language explanations.

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SYSTEM ARCHITECTURE

Three-Tier Architecture

Clean separation of concerns between frontend presentation, API orchestration, and the forensic analysis engine.

Component 01

Frontend Application

React · Vite · Tailwind CSS

Netlify

Web UI for document uploads, real-time analysis progress, visual overlay rendering, and interactive report review. Polls backend job endpoints for status and retrieves JSON report + overlay assets.

- Secure file upload with drag-and-drop
- Progress polling and status display
- Interactive document overlay viewer
- Report download (PDF / JSON)

Component 02

Backend API

Python · FastAPI · Firebase

Render / Docker

Handles document ingestion, job orchestration, and authentication. Manages encrypted file storage in Firebase Cloud Storage, issues analysis jobs, and returns structured results with overlay data.

- Document upload & encrypted storage
- Async job queue & status tracking
- Firebase Auth + role-based access
- Structured JSON report serving

Component 03

Forensic Analysis Pipeline

OpenCV · Tesseract · NumPy · Pillow

Docker container

Modular analysis engine with six independent stages. Each module produces a normalized score and potential anomaly signals. A weighted risk engine aggregates all results into the final authenticity score.

- OCR extraction & text structure parsing
- Pixel forensics & artifact detection
- Layout & font consistency checks
- Metadata forensics & risk aggregation

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HOW IT WORKS

End-to-End Verification Flow

From upload to final report in seconds. Each step is audited and every decision is explainable.

1. **User uploads a document** — 01
   Accepts PDF, JPG, or PNG via drag-and-drop or file picker.
2. **Secure encrypted storage** — 02
   Backend stores the document encrypted at rest in Firebase Cloud Storage, scoped to the user's UID.
3. **Forensic pipeline triggered** — 03
   The document enters the six-module analysis pipeline. Preprocessing renders PDF pages and normalizes images.
4. **Multi-module analysis** — 04
   OCR extraction, pixel anomaly detection, layout analysis, font consistency checks, and metadata inspection run in sequence.
5. **Risk scoring & aggregation** — 05
   Each module produces a normalized score. The risk engine applies weighted aggregation to compute the final authenticity score.
6. **Visual overlay rendering** — 06
   Frontend receives overlay JSON and renders highlighted suspicious regions on the document canvas.

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screensho

Mark-up and sh

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TECHNICAL PIPELINE

Seven-Stage Analysis Engine

Each module is independently testable and produces normalized scores. Click any stage to inspect its description and sample output.

01 **Document Preprocessing** PyMuPDF Pillow NumPy

02 **OCR Extraction** Tesseract pytesseract OpenCV

03 **Pixel Forensics** OpenCV NumPy SciPy

04 **Layout Consistency** OpenCV Pillow heuristics

05 **Font Consistency** fonttools PDFMiner heuristics

06 **Metadata Forensics** ExifRead PyPDF2 python-dateutil

07 **Risk Scoring Engine** NumPy custom weights FastAPI

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Forensica AI

Problem | Solution | Architecture | Pipeline | Tech Stack | View Demo

EXPLAINABLE AI OUTPUT

Every Flag Is Fully Explained

Verification officers see not just a risk score, but exactly which region triggered a flag, which module detected it, and a plain-language explanation — enabling confident, auditable decisions.

Anomaly Explanations (Sample Report)

Font Inconsistency — HIGH

GPA field uses a different typeface (Helvetica Neue) than the rest of the certificate (Times New Roman). Likely text replacement.

Module: Font Analysis Confidence: 94% Region: (18%, 52%)

Pixel Noise Anomaly — HIGH

JPEG compression artifacts around the student name differ significantly from surrounding regions — indicative of patch editing.

Module: Pixel Forensics Confidence: 89% Region: (15%, 32%)

Metadata Mismatch — MEDIUM

EXIF creation date (2024-03-10) conflicts with claimed issue date (2026-01-15). Document was likely created before the stated date.

Module: Metadata Forensics Confidence: 76%

Layout Alignment Drift — LOW

Structured JSON Report

Download

report.json – Forensica AI


```
{
  "document_id": "doc_7f3k2m9x",
  "risk_level": "high",
  "authenticity_score": 34,
  "anomalies": [
    {
      "id": "ano_001",
      "type": "Font Inconsistency",
      "severity": "high",
      "confidence": 0.94,
      "module": "Font Analysis",
      "region": { "x": 18, "y": 52, "w": 32, "h": 8 },
      "explanation": "GPA field uses Helvetica Neue, diverging from expected Times
    ],
```


Why Explainability Matters

Traditional black-box systems only return a pass/fail verdict. Forensica AI tells verification officers *exactly* which region triggered a flag, which module detected it, and why — enabling confident, reproducible, and legally defensible decisions.

Full Verification Report

Download the complete PDF verification report for the sample document, including risk

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TECH STACK

Production-Grade Technology

Built on proven, open-source libraries with a clear path from prototype to production deployment.

Frontend

HTML5 | CSS3 | Vanilla JavaScript | Fetch API | Canvas API

Zero-dependency UI for maximum compatibility and fast load times

Backend API

Python 3.11 | FastAPI | Uvicorn | Pydantic | asyncio

Async-first REST API with automatic OpenAPI documentation

CV & Analysis

OpenCV 4.x | NumPy | Pillow (PIL) | PyMuPDF (fitz) | Tesseract OCR | pytesseract | SciPy

Battle-tested computer vision stack for image forensics

Security & Storage

Firebase Auth | Firestore | Firebase Cloud Storage | JWT tokens | AES-256 encryption

Role-based access control with encrypted document storage

Deployment

Docker | Docker Compose | Netlify (frontend) | Render (backend) | GitHub Actions CI

Hybrid local + cloud deployment with containerized backend

DEPLOYMENT ARCHITECTURE

Netlify CDN → FastAPI (Render) → Pipeline (Docker) → Firebase Storage

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Forensica AI

Problem Solution Architecture Pipeline Tech Stack

View Demo

SECURITY & STORAGE

Security-First by Design

Sensitive documents deserve serious protection. Every layer of Forensica AI applies defense-in-depth security principles.

Encryption at Rest

All uploaded documents are encrypted with AES-256 before storage in Firebase Cloud Storage. Encryption keys are managed per user UID.

Auth Token Verification

Every API request requires a valid Firebase Auth JWT token. Document access is scoped strictly to the owning user.

Role-Based Access

Judges and demonstrators are granted read-only access to pre-approved document sets via Firebase custom claims and Firestore security rules.

User-Scoped Isolation

Documents and analysis results are stored under user-scoped Firestore collections. Cross-user access is blocked at the database rule level.

Audit Log

Every document access and analysis run is logged with timestamp, user UID, action type, and IP address for full traceability.

Ephemeral Processing

Documents are processed in memory-isolated containers. Raw file data is not logged or persisted beyond the encrypted storage layer.

Deployment Strategy

Hybrid architecture supporting both local and cloud deployments

Local Demo

Recommended for judging

Run entirely on-device. No external network calls. Best for hackathon demos.

Cloud Deployment

Production ready

Dockerized backend on Render, frontend on Netlify, Firebase for auth and storage.

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

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Dockerized backend on Render, frontend on Netlify, Firebase for auth and storage.

AES-256 at-rest encryption | **JWT Auth** every request | **RBAC** role-based access | **Audit Log** full traceability

PERFORMANCE & RESULTS

Validated on Internal Test Sets

Sample test results are included in the downloadable project brief for judge evaluation and reproducibility.

95%
Detection Rate
Detected 95% of synthetic edits in internal test set

92%
Precision
False positive rate of 8% on clean document set

2.3s
Avg Analysis Time
Full 6-module analysis on standard certificate PDF

6
Analysis Modules
Independent, testable forensic analysis stages

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share